

Dr. Jacob W. Archambault

AWS AND AZURE-CERTIFIED DEVOPS DEVELOPER WITH SECRET SECURITY CLEARANCE

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EDUCATION

2017	Ph.D. , philosophy	FORDHAM UNIVERSITY, BRONX, NY
	Areas of specialization: logic, semantics, medieval philosophy. Dissertation: <i>The development of the medieval Parisian account of formal consequence</i> . Director: Professor Gyula Klima. Passed without corrections.	
2011	M.A. , philosophy (cum laude)	UNIVERSITY OF HOUSTON, HOUSTON, TX
2009	B.A. , philosophy (cum laude)	FRANCISCAN UNIVERSITY OF STEUBENVILLE, STEUBENVILLE, OH
2021	A.S. , software development	JEFFERSON COMMUNITY AND TECHNICAL COLLEGE, LOUISVILLE, KY

SECURITY CLEARANCE AND PROFESSIONAL CERTIFICATIONS

Mar 2025	DoD Security Clearance - Secret
Oct 2025	LPI - Linux Essentials
Oct 2025-Oct 2028	CompTIA Linux+ certified CE
Dec 2024-Dec 2027	CompTIA Security+ certified CE
Dec 2024-Dec 2025	Microsoft certified: Azure developer associate
July 2024	PCEP - Certified Entry-Level Python Programmer
March 2024	Microsoft certified: Azure fundamentals
Jan 2024-Jan 2027	AWS certified developer - associate
Dec 2023-Dec 2026	AWS certified solutions architect - associate
Jul 2023-Jan 2027	AWS certified cloud practitioner

EMPLOYMENT

	SAIC	
June 2025-Present	Sr. Software Engineer	CLOUD COMPUTING TEAM
	Architected and automated Varonis-Virtru integration for decrypting Virtru Trusted Document Format (TDF) base-64 encrypted DoD SEC//REL classification and releasability attributes from Virtru JSON manifests and making those attributes visible in Varonis data security platform on a daily cron schedule using Varonis scanning, scheduled SSRS reports, and Powershell scheduled tasks. Implemented front-end redesign of React portal application deployed on AWS Elastic Kubernetes Service and configured with FluxCD. Assisted in updating GitLab pipelines to handle securely passing token credentials and SSL certificate to Docker build for accessing private package repository. Installed and configured AWS CloudWatch agents across over 100 Linux and Windows EC2 instances with AWS Systems Manager for improved monitoring and visibility. Wrote Confluence documentation on deploying, configuring, and integrating Varonis with Active Directory, SSRS, Exchange, Virtru and SharePoint.	
Oct-June 2025	Software Engineer	
	Using Packer, Ansible, and Terraform infrastructure as code, automated deployment of custom STIG-compliant Windows Server 2022 Amazon Machine Image (AMI), AWS Route53 records, security groups, and EC2 instances for distributed system hosting Varonis data security platform, SQL Server, and SOLR/Zookeeper software into secure DoD environment, with AWS user data PowerShell scripts for automating SQL Server install and configuration, creating shared directories, installing Windows features, opening firewall ports, and ensuring Service Principal name assignment for obtaining Kerberos tickets. Assisted client in adding Dynamically Adapting to Application Sizes (DATAS) feature to Azure App Service Blazor .NET 8 application to mitigate degraded performance while refactoring to resolve root cause IConfigurationBuilder memory leak.	

TATA CONSULTANCY SERVICES

Apr-Oct 2024 Developer TELECOMMUNICATIONS/IVR TEAM

Led technical interviews for new C# .NET hires. Tested and monitored login systems, load balancer health, database disk usage, call quality, website status and other metrics via monitoring dashboards, PowerShell and SQL Server scripts for Avaya voice-over internet protocol (VoIP) phone systems and related infrastructure for automated interactive voice response (IVR) military health care phone services.

VENMINDER

Feb -Nov 2023 Software Developer II PRODUCTION SUPPORT TEAM

Refactored multiple LINQ-to-SQL queries to prevent timeouts for our largest customers and realize major performance improvements and AWS cost savings after companywide SQL Server-to-PostgreSQL migration. Replaced longer monthly retrospectives with shorter more frequent retrospectives aligned with our weekly release cycle, leading to more consistent follow-through and better cross-department collaboration.

May 2022-Feb 2023 CONTROL ASSESSMENTS TEAM

Created DynamoDB tables via AWS CloudFormation, XUnit .NET unit tests, C# immutable records, sealed request and command handlers, and Typescript Aurelia components for control assessments containerized microservice instantiating CQRS architecture running on AWS ECS Fargate, and added Datadog logging and generic batch update ability to our DynamoDB data access library using AWS SDK for .NET.

PETPARTNERS

Oct 2021-May 2022 Associate Software Developer PRODUCTION SUPPORT TEAM

Unified redundant API calls and removed Cartesian explosion from Entity Framework code for premium tracking feature causing slow coverage period loading, leading to a 26X performance improvement. Created Azure DevOps bug template subsequently adopted across entire company, deployed web APIs via Azure DevOps pipelines, and reviewed YAML files for Azure DevOps pipelines updates. Formulated solutions for critical bugs and data fixes in Angular and C# .NET for both customer-facing and internal websites. Implemented and wrote wiki documentation for the company's release process, git workflow, and numerous SQL data fixes.

April-Oct 2021 BACKEND TEAM

Created backend logic, C# web API endpoints, and normalized SQL tables via Entity Framework migrations for handling commission payments to subcontracted companies and internal call center agents for new flagship employer-based pet insurance product. Converted SQL stored procedures for retrieving claims, policy losses, and veterinarian information in monolithic application to Entity Framework backend called by web API microservices, and scripted automation of .NET web API publication for local development environments.

WAYSTAR

May 2020-April 2021 Software Developer Apprentice DEVOPS APPLICATIONS TEAM

Pioneered work converting .NET Framework projects to containerized .NET Core projects running on Docker; co-led company-wide training on using YAML for Azure DevOps pipeline configuration; built, deployed, and configured team projects with Azure DevOps, and set up logging of automated alerts from A10 load balancers to Microsoft Teams and Splunk.

Jan-April 2021 JAVA BOTS TEAM

Fixed bugs in Chrome and Firefox web crawlers obtaining patient payment information from insurance provider websites, written as Java Maven projects using Selenium, working in a trunk-based development workflow with IntelliJ IDEA and Eclipse IDEs, deployed those fixes to development and production environments with Jenkins, and recorded progress on bugs in kanban system using Jira.

Aug 2019-May 2020 ARCHITECTURE TEAM

Contributed to a company-wide full-stack ASP.NET MVC in-house external applicant testing website with Visual Studio and Git version control, including writing C# code, SQL and Entity Framework LINQ-to-SQL queries, and JavaScript/jQuery functions for DOM manipulation and sending/retrieving server information with AJAX for features allowing teams to add or remove team members and restrict admin access to tests and test templates to their owning teams, added RabbitMQ integration to the project for sending emails, and improved HTML/CSS web accessibility and design responsiveness by adding semantic HTML, media queries, and Flexbox layouts.

FORDHAM UNIVERSITY

June 2015- May 2017

Distinguished research fellow

Bronx, NY. Published numerous articles, book contributions, and an edited volume with publishers including Cambridge University Press, Brill and Springer, secured grants and honors from institutions including the University of Salzburg and the University of St Andrews, and presented papers at prestigious institutions across three continents, primarily in the history and philosophy of logic. For a full list of publications and presentations, see my CV.

SELECT PROGRAMMING PROJECTS

Bash

PYTHON POSTGRESQL TOOLBOX

A Dockerfile and shell script for creating a Fedora Toolbx containerized development environment with Python, Pip, PostgreSQL, and VSCodium, the open-source version of Visual Studio Code, pre-installed.

AWS CLI CONTAINERIZED

A .bashrc file allowing the user to easily execute AWS commands using the containerized version of the AWS command line interface (CLI) with Podman.

Python

POSTGRESQL S3 BACKUP

A command line interface (CLI) tool, written in Python with pytest unit testing and built with pdm, for creating and storing PostgreSQL backups either locally or to an AWS S3 bucket. Includes an accompanying bash script for setting up a containerized test postgresql database and storage volume with Podman.

FLASK NOTES

A Python Flask web application using SQL Alchemy allowing users to login, create, read, update, and delete Markdown notes to a PostgreSQL database running on two alpine images with Docker compose.

C#

PERSONAL WEBSITE

A .NET 8 MVC website with links to my programming projects, published papers, resume and CV, deployed to Azure App Service in a Docker Linux container and runnable from a Kubernetes deployment YAML file. Features implemented include partial views and dynamic page content generation, bundling and minification, custom routing and http status code error handling.

AUTO LOT MVC STORE

A three-project .NET 5 MVC solution consisting of a website front-end, a data access layer, and a model layer. Makes use of C# 10 records, razor view components, custom tag helpers, exception filters, custom routing, YAML pipeline configuration with GitHub actions, and LINQ queries via entity framework core to a SQL Server relational database.

RECURSIVE FOLDER COPIER

A .NET 5 file I/O application recursively copying files to an output directory based on user prompts for filters by date, regex pattern, number of files to match per subdirectory, making use of a decorator pattern, dependency injection and multiple implementations of a common interface.

TypeScript, JavaScript, and web development

AURELIA BLOG

An Aurelia TypeScript application implementing a multi-stage build with nginx built on a slimmed-down alpine Docker container image that calls the jsonplaceholder posts API.

ANGULAR STUDENT INFO

An Angular and Node.JS app for updating student midterm and final exam info, retrieved from a MongoDB database via mongoose.

Java

EMPLOYEE REFACTOR

A Java tutorial showing through its git history how to convert a tightly coupled application using inheritance and mutable fields into one implementing best practices including strong encapsulation, composition over inheritance, object cohesion, loose coupling, immutability, and dependency injection, via an application printing out information for various types of employees. Includes an extensive README explaining the implemented design decisions.

SKILLS

AWS	DYNAMODB, CLOUDFORMATION, FARGATE, ECS, CLOUDWATCH, S3, IAM, AWS CLI
DevOps	TERRAFORM, PACKER, KUBERNETES, PODMAN, DOCKER, BASH, POWERSHELL
Microsoft Azure	AZURE DEVOPS PIPELINES, APP SERVICE, AZURE DEVOPS TEMPLATES, AZURE CLI
Python	FLASK, PDM, PIP, PIPENV, PANDAS, MATPLOTLIB, JUPYTER NOTEBOOKS
.NET	BLAZOR, ENTITY FRAMEWORK CORE, ASP.NET CORE, LINQ, XUNIT .NET
Databases	DYNAMODB, POSTGRESQL, SQL SERVER, MONGODB
Web development	TYPESCRIPT, REACT, ANGULAR, AURELIA, NODE.JS, BOOTSTRAP, JQUERY, WORDPRESS
JVM	INTELLIJ IDEA, SPRING, JAVA, SELENIUM, JUNIT, MAVEN, JAVAFX, ECLIPSE